QIRI (Quantum Integrated Research Institute Inc.)
Tokyo 107-0061, Japan
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Editors, *Physical Review Letters*

Dear Editors,

We submit the manuscript "Exact exponential heat rectification from the Liouvillian skin effect" by Hikaru Wakaura and Taiki Tanimae for consideration as a Letter.

Claim. A reciprocal tight-binding chain with directional bond dissipation rectifies heat with a ratio that obeys an exact law, $\delta Q_f/\delta Q_r = \exp[2(L-1)/\xi]$, where ξ is the Liouvillian skin localisation length; the law is independent of energy, temperature and on-site disorder, and a 40-site chain reaches 99% rectification.

Why Physical Review Letters. Thermal diodes have so far required nonlinearity or structural asymmetry and rectify by at most a factor of ten, independent of device size. Our result shows that dissipation alone rectifies heat, that the ratio grows exponentially with length, and that one number controls it. The two works that define the Liouvillian skin effect, Song, Yao and Wang [Phys. Rev. Lett. **123**, 170401 (2019)] and Haga, Nakagawa, Hamazaki and Ueda [Phys. Rev. Lett. **127**, 070402 (2021)], and the cold-atom observation of Liang *et al.* [Phys. Rev. Lett. **129**, 070401 (2022)] appeared in this journal; our Letter answers the transport question they left open, with three further results of general interest: the controlling quantity beyond nearest neighbours is the non-Bloch localisation length and not the spectral winding number, interactions up to $2J$ do not renormalise ξ (matrix-product-state simulations calibrated against the exact free-fermion solution), and a finite-temperature bath closes the entropy balance with a cost per rectified heat that grows as $1/\xi$. The proposal is within reach of momentum-lattice condensates and superconducting circuits with engineered loss.

Suggested referees (no conflict of interest with the authors):

- Zhong Wang, Tsinghua University (non-Hermitian skin effect, chiral damping)
- Masahito Ueda, University of Tokyo (Liouvillian skin effect)
- Aashish A. Clerk, University of Chicago (reservoir engineering, nonreciprocity)
- Henning Schomerus, Lancaster University (transport in non-Hermitian systems)
- Dario Poletti, Singapore University of Technology and Design (boundary-driven open quantum systems)

Previous submissions. Neither this manuscript nor any closely related manuscript has been submitted previously to an APS journal or to any other journal.

Length and format. The Letter is 3715 APS word-equivalents (2875 words of text, five displayed equations, four figures), within the Letter limit; a Supplemental Material of seven pages (derivations, matrix-product-state precision analysis, additional data, experimental parameters) accompanies it.

Preprint and data. The manuscript is posted as arXiv:[number to be inserted at submission]. All data and the code that regenerate every figure are deposited at Zenodo (DOI to be inserted at submission).

Sincerely,

Hikaru Wakaura and Taiki Tanimae